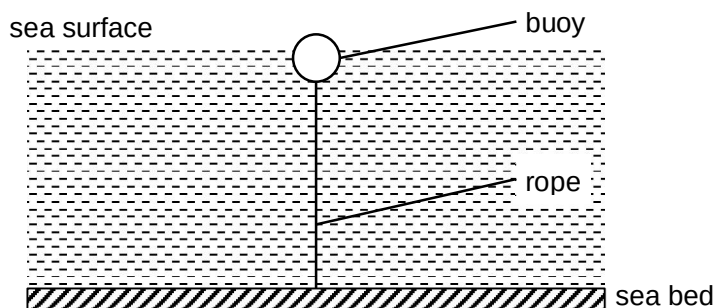


- 1 (a)** A buoy is held partially submerged in sea water by a rope anchored to the sea bed as shown in Fig. 1.1. A fifth of the volume of the buoy is above the sea surface.



**Fig. 1.1**

The buoy has volume  $7.5 \times 10^{-2} \text{ m}^3$  and mass  $8.0 \text{ kg}$ . The mass of the rope may be neglected. The density of sea water is  $1.03 \times 10^3 \text{ kg m}^{-3}$ .

- (i)** Explain what is meant by *upthrust*.

.....  
 .....  
 ..... [1]

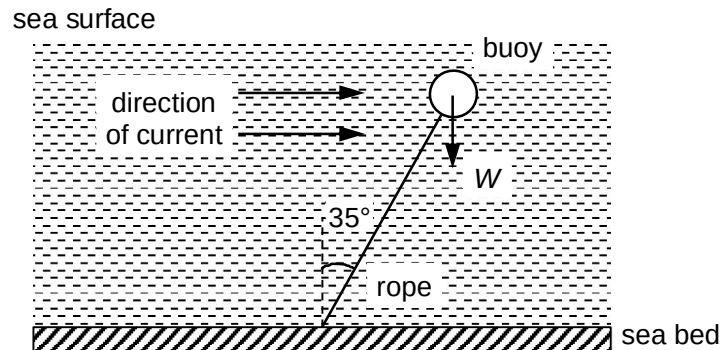
- (ii)** Calculate the value of the upthrust  $U$  on the buoy.

$U = \dots\dots\dots \text{ N}$  [2]

- (iii)** Show that the tension in the rope is  $530 \text{ N}$ .

[1]

- (b) Current in the sea water during high tide cause the buoy in (a) to be displaced so that it is fully submerged and the rope makes an angle of  $35^\circ$  with the vertical, as shown in Fig. 1.2.



**Fig. 1.2** (not to scale)

The buoy may be considered to be acted upon by four forces, tension  $T$  in the rope, a horizontal force  $D$ , upthrust  $U$  and weight of buoy,  $W$ .

- (i) The force  $W$  is shown in Fig. 1.2.

On Fig. 1.2, sketch and label the forces  $T$ ,  $U$  and  $D$ .

[1]

- (ii) By resolution of forces, determine the magnitude of the force  $D$ .

$D = \dots\dots\dots$  N [3]

2 (a) (i) Define *gravitational potential* at a point